TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108
Accredited with NAAC A+ Grade
Approved by AICTE, New Delhi, Govt. of Maharashtra

(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)
Department of Information Technology
Session 2023-24(Even Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

Class Test 1
Semester: VI
Programme: B. Tech Information Technology
Course: Natural Language Processing

Time: 1 hours **Max Marks:** 20 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33608	Course Outcome
CO1	Classify the ability to preprocess text data using NLP techniques.
CO2	Demonstrate syntactic parsing methods, including context-free grammars and dependency parsing
CO3	Analyze semantic structures and apply information retrieval techniques to extract meaningful insights from text.
CO4	Interpret language models and perform text classification using algorithms in NLP.
CO5	Evaluate real-world applications of Natural Language Processing

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	What is tokenization?	CO1	2	2
b.	Define parsing in NLP.	CO2	2	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain normalization and noise removal.	CO1	2	4
b.	Describe Word2Vec and GloVe.	CO1	2	4
c.	Explain FastText and contextual embeddings (BERT).	C01	2	4

Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain dependency parsing.	CO2	2	4
b.	Describe chunking in NLP.	CO2	3	4
c.	Compare rule-based and statistical POS tagging.	CO2	3	4

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create